PlantTraitNet: An Uncertainty-Aware Multimodal Framework for Global-Scale Plant Trait Inference from Citizen Science Data

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Lecture Transcript

Hello, everyone. So today I'm presenting our work, Plant Trait Net, an uncertainty of the Multimodal Framework for Global-scale Plantrate Inference from Citizen Science Data. This work was done with the help of many collaborators at the University of Pri and many more around the globe.

Um,

So from the title established that we are trying here to get global scale plant map, but what is the motivation behind it? What are we, why are we doing it? Well, that's the first thing on our agenda today. So plant rates such as plant height, leaf area, leaf nitrogen, they govern the growth and productivity of the plant. So how a taller plant has more access to sunlight. If a plant has more leaf nitrogen, it can be more productive. It also helps in carbon storage. And ultimately these plant traits, they regulate climate by regulating the carbon cycle and nutrient cycle.

So these uh global plant rate map, they play, they play an important role to understand ecosystem functioning and earth system modeling. But to have global plant rate map, we need a data which is globally available and therefore we resort to citizen science contribution. There's a platform called Ionatist Dataset, which is like an Instagram for plant images. People who are nature enthusiasts, they go out and they capture plant images, and you get corresponding species label on this plant. As you see, this data is super abundant over here.

But this data set doesn't have one on one correspondence between plant photo and trait estimation.

So we bring ri database into the picture. Tri Database is another scientific database where scientists across the globe, and they want to measure or they want to understand a species better, they get trait measurements, multiple occurrences across the globe. So this data set provide us trait distribution per species.

We use this. Data set,

try to establish one on one correspondence between planned photographs from Ayatollah.

And corresponding plant rates in tri using species as as a bridge.

More specifically, from tribe per species you have a distribution of, uh, you get a distribution of traits, and every epoch we sample a label from this distribution and annotate this image. This gives a weak, noisy annotation, but it also captures the inherent trait variability.

But how do we assess it? So we have super noisy, weak annotation to assess it, we uses plot open. Splot open is again a global data set of vegetation plots

which where you have a species composition. So you scientists go out in the field, they understand the species composition on that field, and then again use this tri database as its trade distribution to get community weighted trait mean. Weighted by occurrences of a species on that field, it is spatially very sparse, but it's highly reliable. There's no other database that has global plant image to trade correspondence, so we have to use this data set for assessment and try database as weak label annotation. So there's an overview. We have IN data set which is, which has georeferenced plant images. Tri data set gives us plant trait data distribution. And then we use S plot opens communitivated trait value to assess the quality of model protection. Now how do we model it? To model it, uh, we use frozen image encoders. Um, uh, we tried many, and for state of the art we use Dyno V2. We also use depth prior to get the geometric information, uh, spatial cues, um, and we also use geo foundation model. We use Climplicit, which is trained on climate variables so that the embedding gives you the climate information of a location. We use this one dimensional embeddings, we concatenate them and pass them on to multimodal backboard, uh, which then goes to separate four independent nonlinear trait heads, one for each trait that we're predicting here. We're predicting plant height, leaf area, specific leaf area which is leaf weight or leaf mass, and leaf nitrogen, which indicates how much nitrogen a leaf has. But, uh, because the data that we use as citizen science data, it's super noisy, we incorporated uncertainty into the picture. We do not just predict a point value, but we also predict log variance, and high variance means high uncertainty. We train it using negative log likelihood, which you see here. M is for M denotes each trait, N is number of sample. Mumu here is your point prediction. And this S that you see is your standard deviation, uh, that you get from this variance. So we formulate it, formulate this as Gaussian distribution, and the loss is inversely proportional to uncertainty, so dynamically downweight on certain samples.

Now coming to the methodology, although we see that loss dynamically down sample these uncertain samples, if you train them for long, the model learns the noise, it loses its generalizability, uh, generalizability, and therefore we need to handle that. For that we propose uncertainty of their framework. But first, let's understand what kind of noise do we have. So here on the right you see plant images which are highly uncertain for height, um. And you see that there are a lot of images with winter landscape

where you do not see a tree, which which are definitely bad images to infer plant hide from. Another trait, leaf, a specific leaf area which is a leaf-based trait, you see high uncertainty for bare branches, bird flowers where there are no leaves. You see higher uncertainty for leaf nitrogen here, for fuzzy, blurry images because leaf nitrogen can easily be or can be inferred from greenness of the image, but if the image is blurry, the model is super uncertain. So we see that the model predicts task-based or trait-based uncertainty as well. So this uncertainty prediction is reliable, and then we use this to filter out noisy data.

So we iteratively train our model and filter out these noisy data, uh, with high, uh, with high predictive uncertainty, uh, and we do it up to a certain threshold. But is this enough? No, it is not. Uh, there's a limitation which is heterosyasticity. So for taller plants, the residual is in meters as compared to, uh, grasses which would be probably in some meters. So the uncertainty for them would always be higher. So because

of this, we proposed this method, residual of their filtering. Residual of their filtering also help us with the noisy label. You'll see how. So we, we have seen that we use trade distribution to weakly annotate the images, but the, the trade label come from scientific database. However, iNaturalist has plant images from people who probably put it up on their whims, on their liking. So we have immature growth species as well,

growth immature plant plant images as well, which deviate quite a lot from their distributions true value. For example, these images belong to the same species, but this one is the developed plant, and these are. Not, um, and you see where the uncertainty is high and the residual is also high, the plant is super immature, so the residual is higher we remove these samples so that we do not have such noisy weak label impact on the training. And here we show the topmost sample that we removed where the, the aggregation, the height average is super high, up to a few meters, and the predictions are very low because the plant images are also immature.

Now we have a defined data, we do it for certain nation again to reach a threshold. Then we have a refined data, we then train our model. During inference, we aggregate these planned prediction using geo uh information. Into a one degree resolution spatial map. And we compare this 1 degree spatial map against this Splot open data map that we derived using Tri database. It's actually provided by Splot open.

So there's an overview. We do uncertainty-based filtering, residual of the filtering. We train on refined data and evaluated on vegetation survey data.

Now coming to some results, so we show here Plant traitNet, which is trained on raw data, PlantrateNet trained on refined data. We show for all four traits, height, leaf area, specific leaf area, leaf nitrogen. Uh, we report R square, which captures variance, normalize mean absolute error, uh, and Pearson R correlation. We observe here that Plant rate net train on refined data. Usually performs better for most of the traits when trained on than when trained on raw data. We also see that for most of the previous baselines, which were mostly remote sensing focused, or a few over here which were citizen science data focused but didn't include image information but rather just tabular data,

we outperformed them. Except for nitrogen when we didn't perform this, and this, this also uses the baseline also uses citizen science, um, uh, citizen science method but uses a single task model,

single independent model for each trait, which is also computationally computationally expensive. Um, and here we show phylogenetic modulation. So we group related species into group and see the residual error to see if there's systematic bias toward one particular species. For each of the traits we report here residual error. So the idea is to see if if there's a particular bias toward one that we do not observe here. We see that it's randomly distributed. We all, you might also argue that, OK, for some species, the residual is higher, but for some, no, not, but that is attributed again to the heterostatistic nature, where the residual is always higher for trees as compared to grass or uh sher species.

And here we show intraspecies variation, so we do remove immature plant species during training, uh, but we do an inference on them, uh, and we see that the model can nicely capture the intraspecies variation between different intertrade variation, different between different

mature forms of these plants.

Um, OK. And now coming to the ablation. So in this ablation, we show that, um,

Image encoder being used alone

is being outperformed when we use image encoder and geoencoder, especiallylimplicit for geoencoder we

use other foundation model as well, such as set clip and geoclip. Um, but together Dyno V2, Climplicit, and Depthing V2 frozen encoder, we observed that this setting is most robust for most of the trade. We also train a single trade variant and observe that performance usually improves for all trades when we have a single trade variant, but again it's 4 times more expensive.

Now coming to the conclusion, we see that citizen science data here paired with labels from scientific data can infer functional databases, can infer functional traits from images. Uh, we see uncertainty of deep learning, it helps with noisy citizen science images. We see that our model generalize across species and then captures intraspecies variation, and we outperform previous work for most traits on global benchmark, uh, which shows its potential. And here's the uh link you are code for open, uh, for the page which will take you to open source database, open source code, and we thank BFG IFLs and try to support us. Thank you.

Any questions?

Hi, thank you so much for your talk. Um, my question is about the initial part of your framework where you do that weak linking between species across the two frameworks. Um, do you do uncertainty-based filtering at that point, so if the citizen has completely labeled the image as the whole wrong species altogether, is that captured in that uncertainty aware framework, or is that only for the traits that you do that uncertainty aware component? No, we do not do that because. The task is more about trade prediction. We do not want to see if a species, like we trust Iatist framework. Iatist platform is quite an established database, so we want to, uh, we pretty much trust the prediction there. So we do not verify the species are classified as right or not.

Um, hello. Hi, thanks for the really interesting presentation. Um, I'm wondering for like, um, predictors, what's the reason you guys use like embeddings like sad clip from, you know, foundational models uh rather than using like, for example, remote sensing data or

Uh, like images, satellite imagery, like what's the reason for choosing the embeddings rather than those? Yeah, thank you. Uh, so we started with remote sensing data. We use World Clem, Soilrid, uh, Modi's, um, we used all these data and we saw that we do not like, we have to impute a lot. There's a lot of data sparsity in some of the database and Uh, these models, they just can interpolate to unseen location really well, the location for which we do not have the data. So that's why we use geo-encoder. Uh, about remote sensing images, actually that clip is trained on remote sensing data, and so we thought that it would be useful, but we saw that cliplicit clearly outperforms it. And also it established in ecology, uh, ecological space that climates play a huge, a significant role in shaping plant traits. Uh, so that was also demonstrated in our ablation study. So it's mainly like because data scarcity, is it, or?

So just as a follow-up, just for clarity, like is it because more like data scarcity because I mean I understand that that clip is from uh world like world claim and also elevation, all that, but like I was wondering whether Is it because like you have to impute a lot that you use said quid, but I mean, or are there other reasons in terms of like performance comparisons and, you know, other comparisons that you choose? Yeah, it would be nice to try it out, but because we were running into a lot of, uh, problem about interpolation or imputation that we had to do, uh, we thought that using geo-foundation model, which are continuous, which give you, which can generalize or they demonstrate in the papers that they do, we use them.

Thank you very much. Any last question?

And thank you very much.